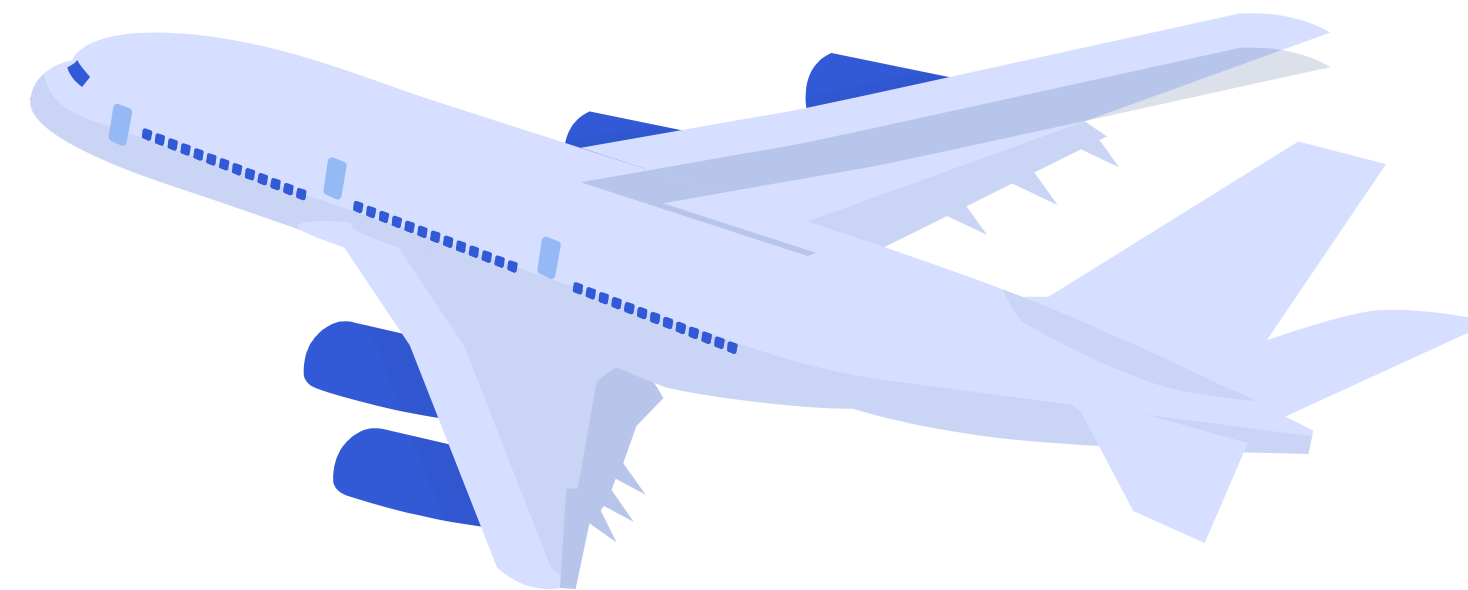
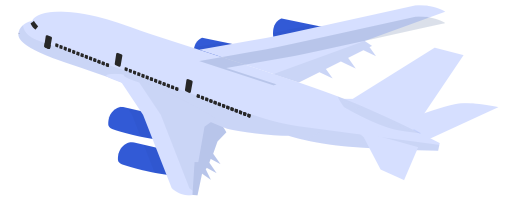




# *Total Passengers Flights Analysis*

Singelyn Center for Innovative Analytics  
MISSA Data Analytics Team





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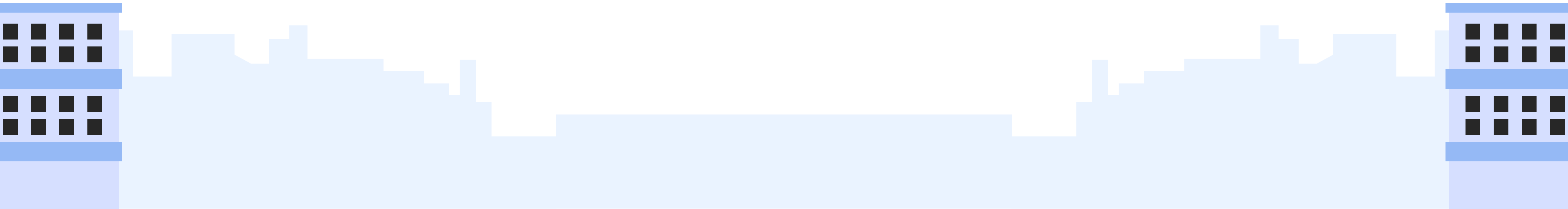
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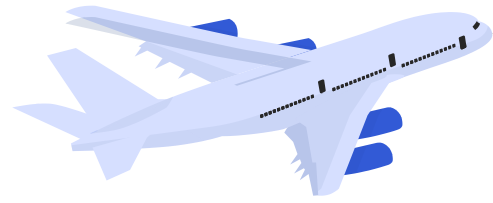
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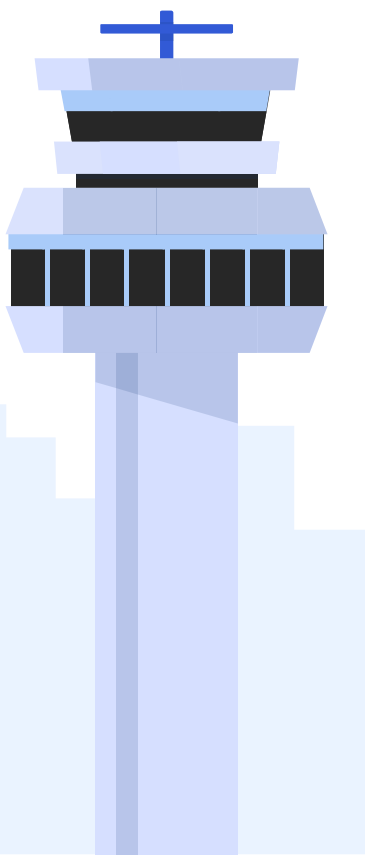
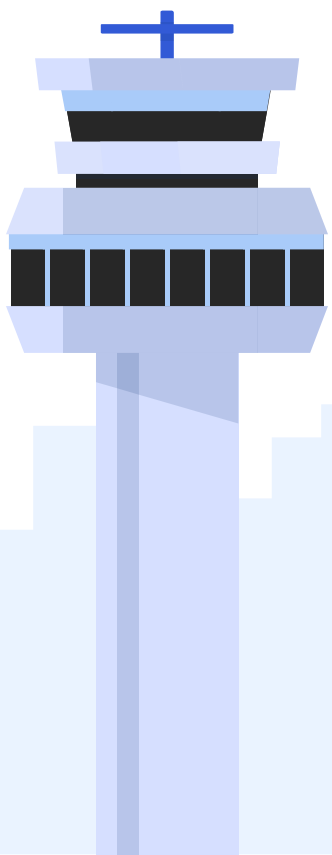
**Future Work**





1

# *Introduction & Project Overview*





# *Project Overview*

This project aims to determine the total number of passengers who flew on Boeing airplanes during 2024, focusing exclusively on revenue-generating commercial flights. It includes all Boeing and McDonnell Douglas heritage models and sub-models

The background features a stylized airport scene. At the top, there are three blue, fluffy clouds. In the upper left, a white commercial airplane with blue accents is flying towards the right. At the bottom, there are two air traffic control towers with blue and white sections and a cross on top. The base of the towers and the area between them are filled with a light blue, pixelated cityscape silhouette.

2

# ***Data Collection & Cleaning Process***



# Data Exploration

## Premium Data Providers Considered



*Limited value in regards to scope of project*

---

## Data Provider Used



BTS is publicly available, structured and standardized, and provides comprehensive coverage on 2024 data spanning a full calendar year

# Data Exploration: **BTS** Statistics



Contains both domestic and international data from US and foreign air carriers

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Inclusive of all airplane models (Boeing, Airbus, Cessna, etc.)

---



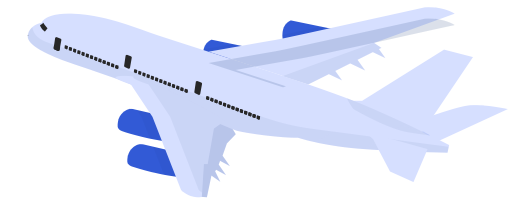
Flights in foreign countries with both origin and destination are **not included**

---



Broad range of data from 1990 to 2025

# Data Cleaning



## Step 1

Merged with aircraft reference sheet to decipher coded data for

- Aircraft type
- Aircraft configuration
- Aircraft group

## Step 2

Built and joined our own aircraft model dataset to add additional columns based on aircraft model:

- Length, Height, Wingspan
- Range
- Engine
- Max Occupancy
- Manufacturer

## Step 3

Our dataset is aggregated, one row summarizes all flights that share the same:

- Aircraft carrier
- Route (origin → destination)
- Aircraft Type
- Month & Year

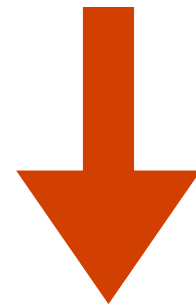
## Step 4

Final Data Preparation:  
We standardized Month format as string > Replaced null with Unknown in origin/destination states > Removed flights with 0 passengers to focus on activity



# Dataset Information

**548,633 x 22**



**458,017 x 33**

Data Types:

DEPARTURES\_SCHEDULED  
Number of Flights  
SEATS  
PASSENGERS  
DISTANCE  
AIR\_TIME  
UNIQUE\_CARRIER  
UNIQUE\_CARRIER\_NAME  
CARRIER\_NAME  
ORIGIN\_AIRPORT\_ID  
ORIGIN  
ORIGIN\_STATE\_NM  
DEST\_AIRPORT\_ID  
DEST  
DEST\_STATE\_NM  
DEST\_COUNTRY

AIRCRAFT\_GROUP  
AIRCRAFT\_TYPE  
AIRCRAFT\_CONFIG  
YEAR  
QUARTER  
MONTH  
Code  
Description  
Avg Passengers per Flight  
Model  
Max Occupancy  
Range (nmi)  
Length(m)  
Wingspan (m)  
Height (m)  
Engine  
Manufacturer

***1 row ≠ 1 flight***

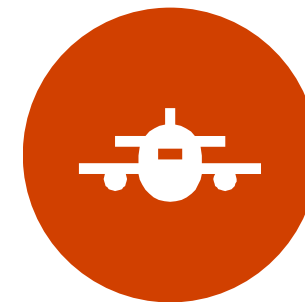
*Each row can represent a # of flights*

# Key Dataset Columns



## ***Passengers***

Number of passengers who actually flew



## ***Model***

Specific aircraft model used (e.g., Boeing 757-200)



## ***Number of Flights***

Total flights recorded for each entry



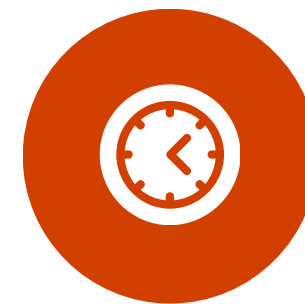
## ***Origin/Destination State***

U.S. states where flights began and ended



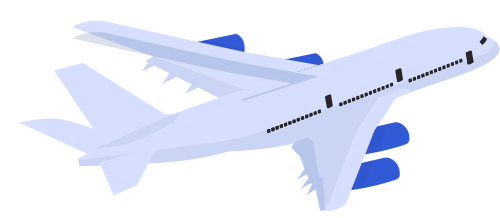
## ***Manufacturer***

The company that built the aircraft (e.g., Boeing)



## ***Month/Quarter/Year***

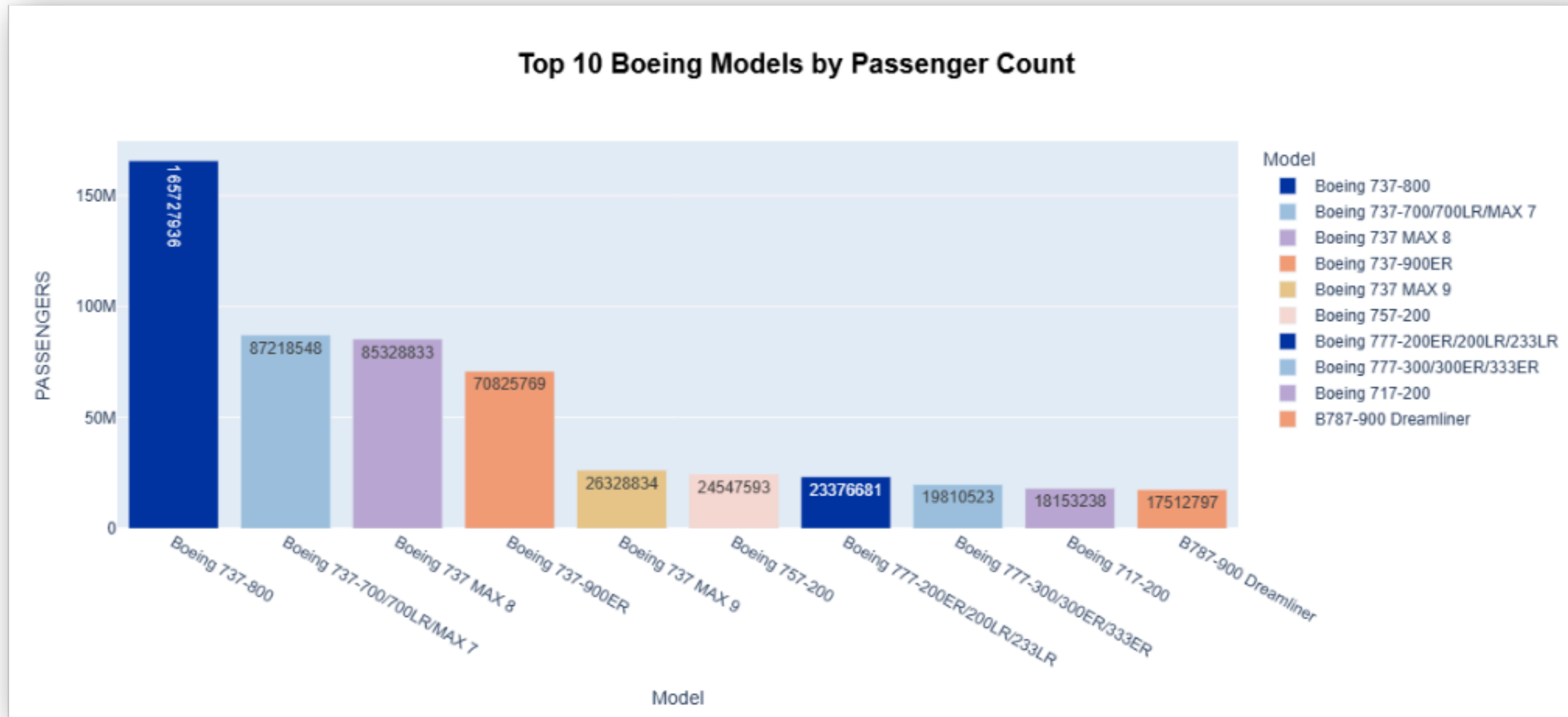
Time period when the flights occurred



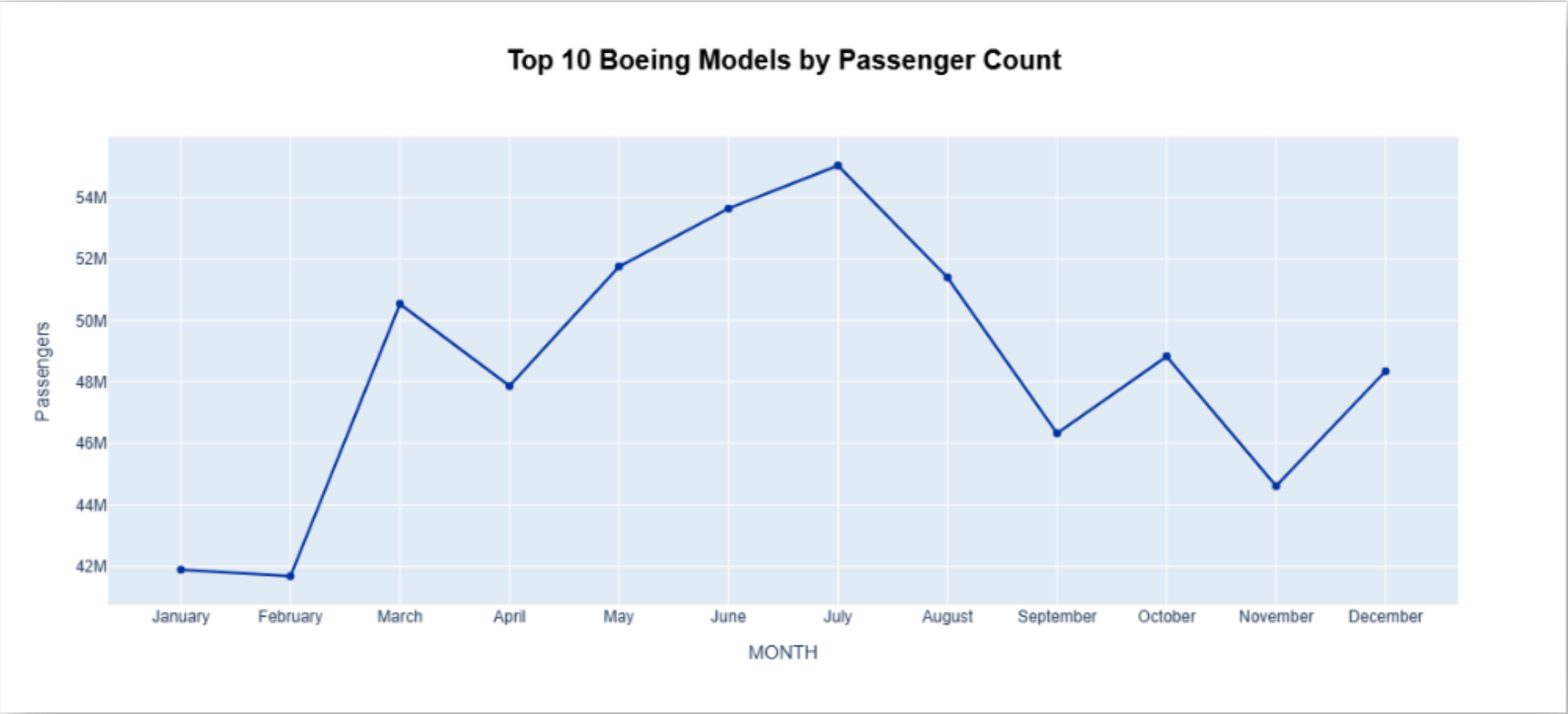
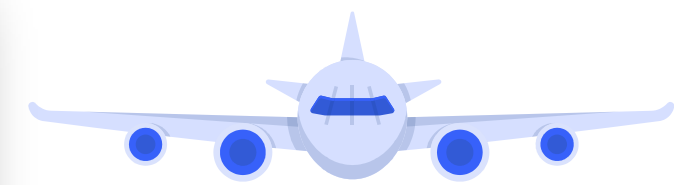
3

# *Visualizations & Dashboard Insights*





The Boeing 737-800 is the most used aircraft, carrying nearly twice as many passengers as the next model, with all top five spots occupied by planes from the Boeing 737 family

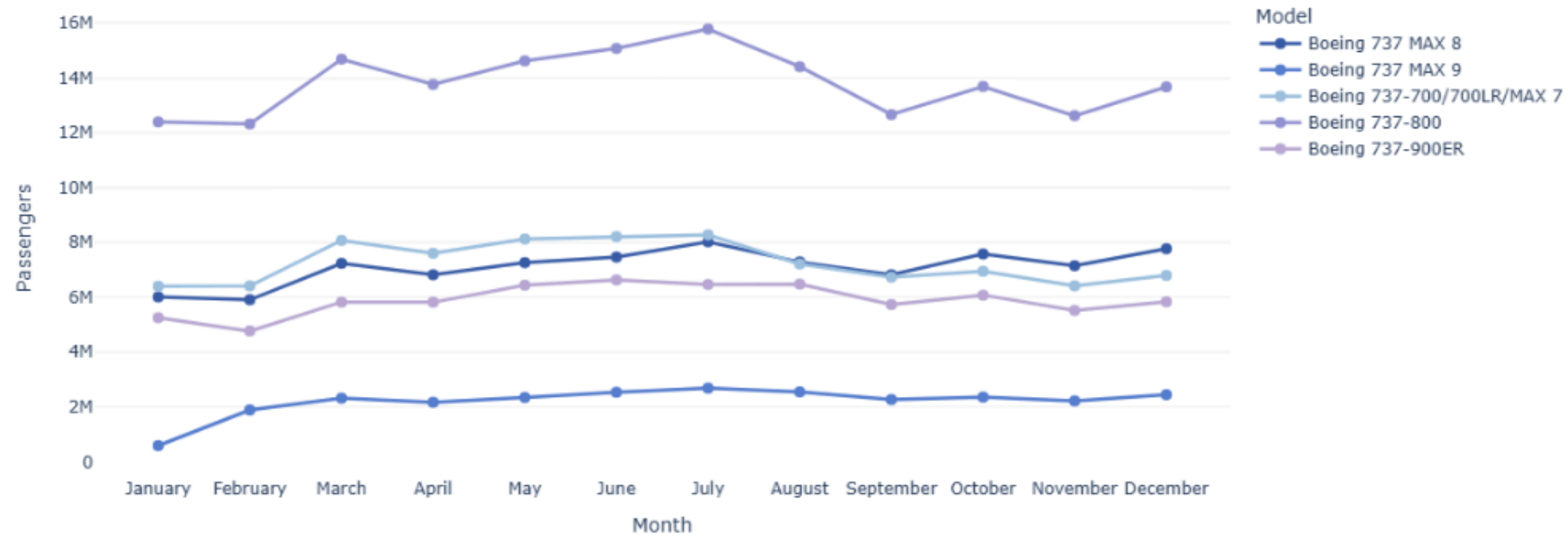


Overall passenger volume for Boeing models peaks in July at around 55 million, with significant declines observed in February and November

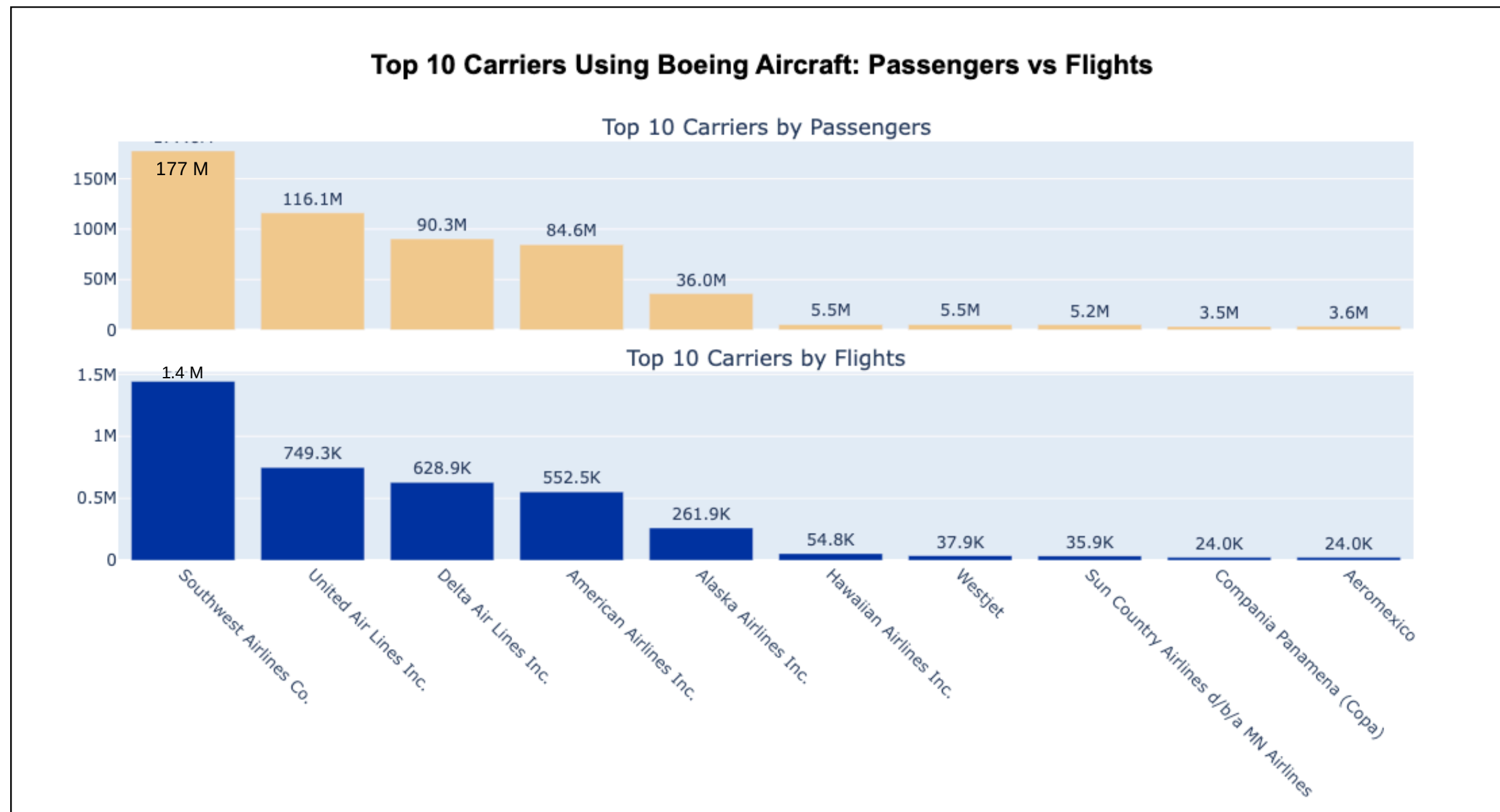
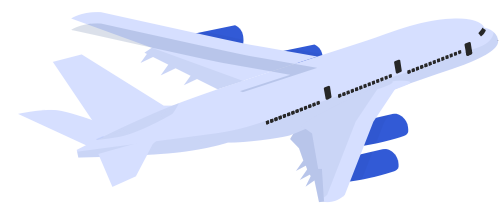




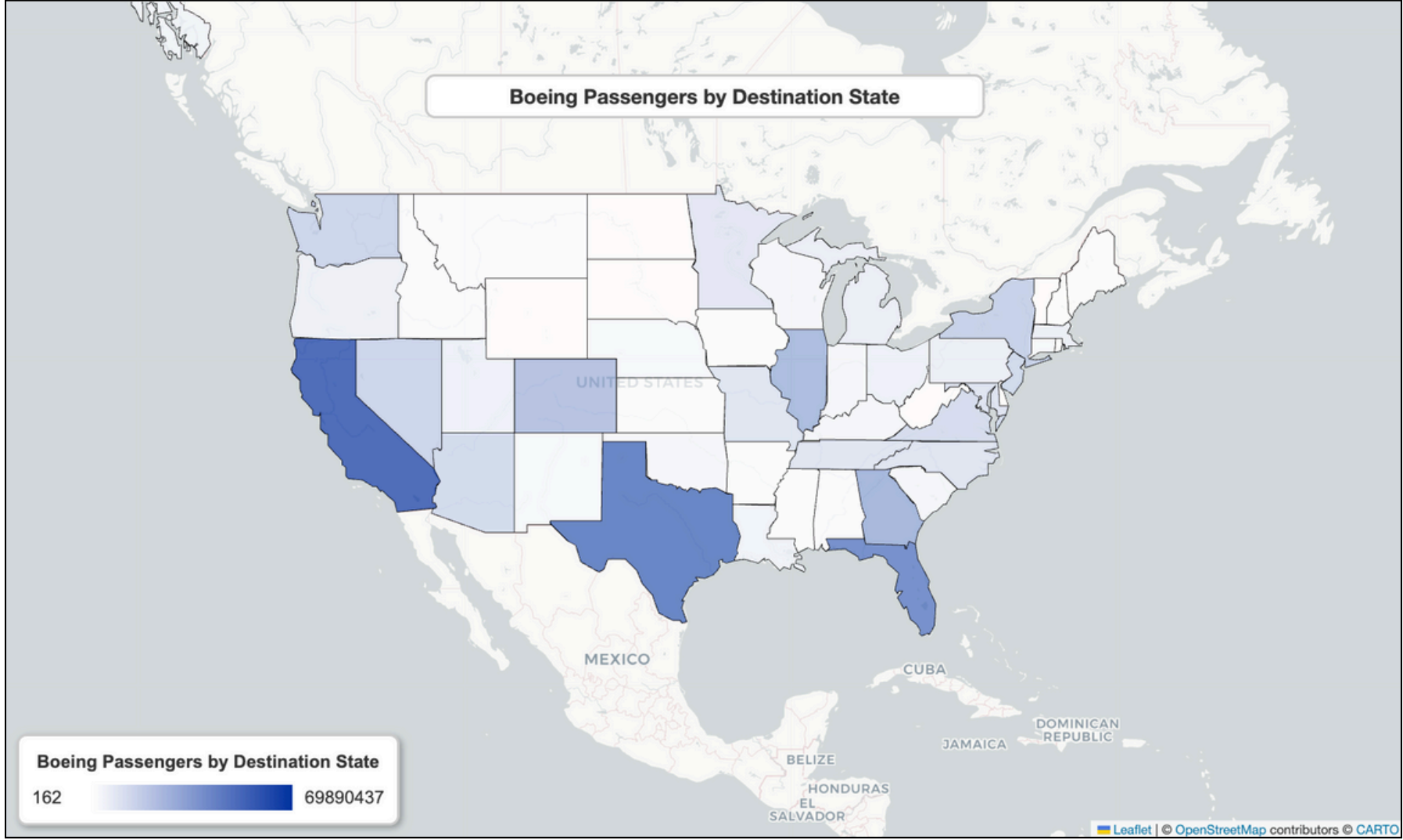
Monthly Passenger Trends for Top 5 Aircraft Models



The Boeing 737-800 consistently leads in monthly passenger counts, with seasonal dips appearing in August and November across most models



For Boeing flights, Southwest operates the most, while Delta, United, and American Airlines move many passengers with fewer flights

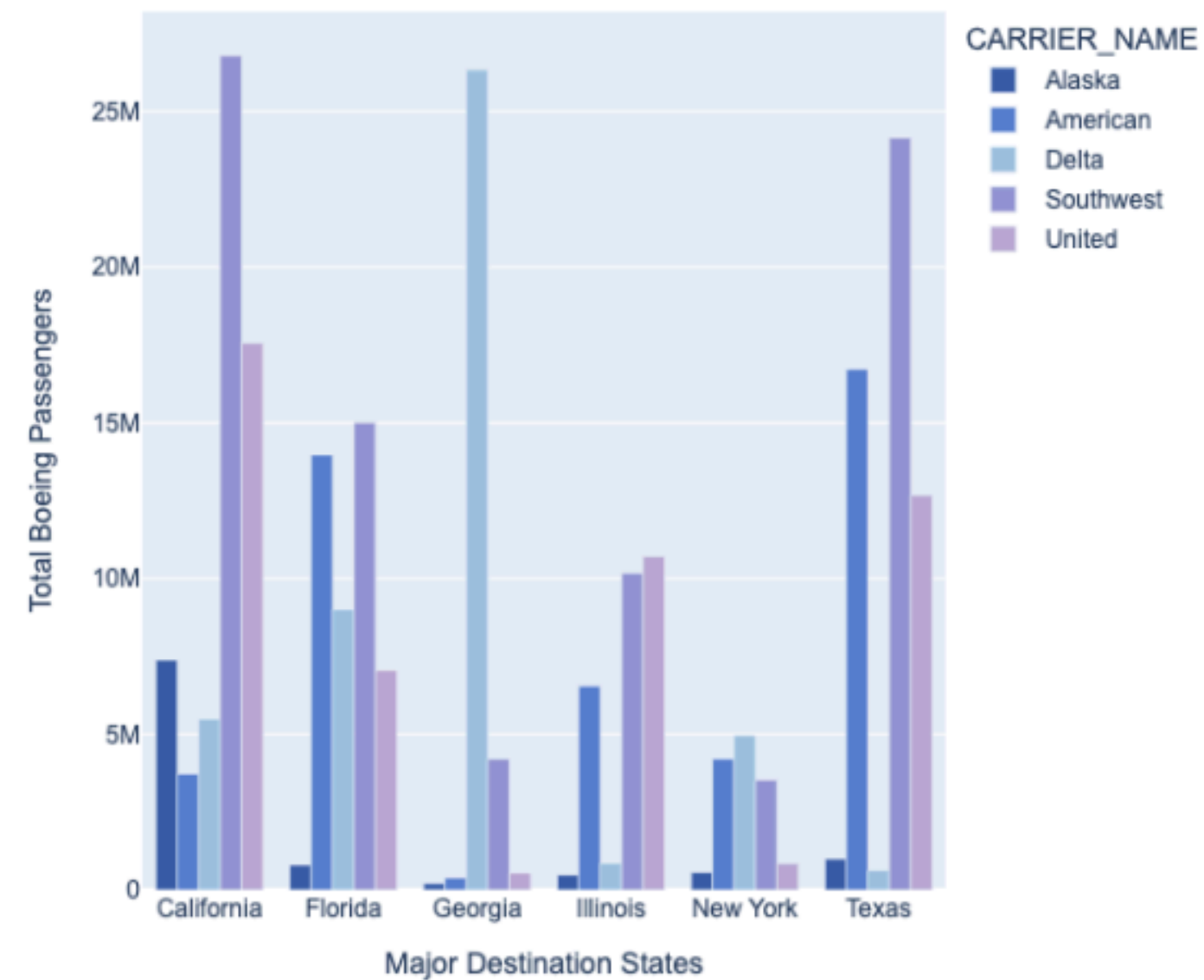


| State      | Passengers |
|------------|------------|
| California | 69,890,437 |
| Texas      | 57,173,442 |
| Florida    | 51,645,254 |
| Georgia    | 32,739,201 |
| Illinois   | 30,847,664 |
| Colorado   | 26,948,195 |
| New York   | 19,745,401 |
| Washington | 18,291,064 |
| Nevada     | 18,132,971 |
| New Jersey | 16,438,414 |

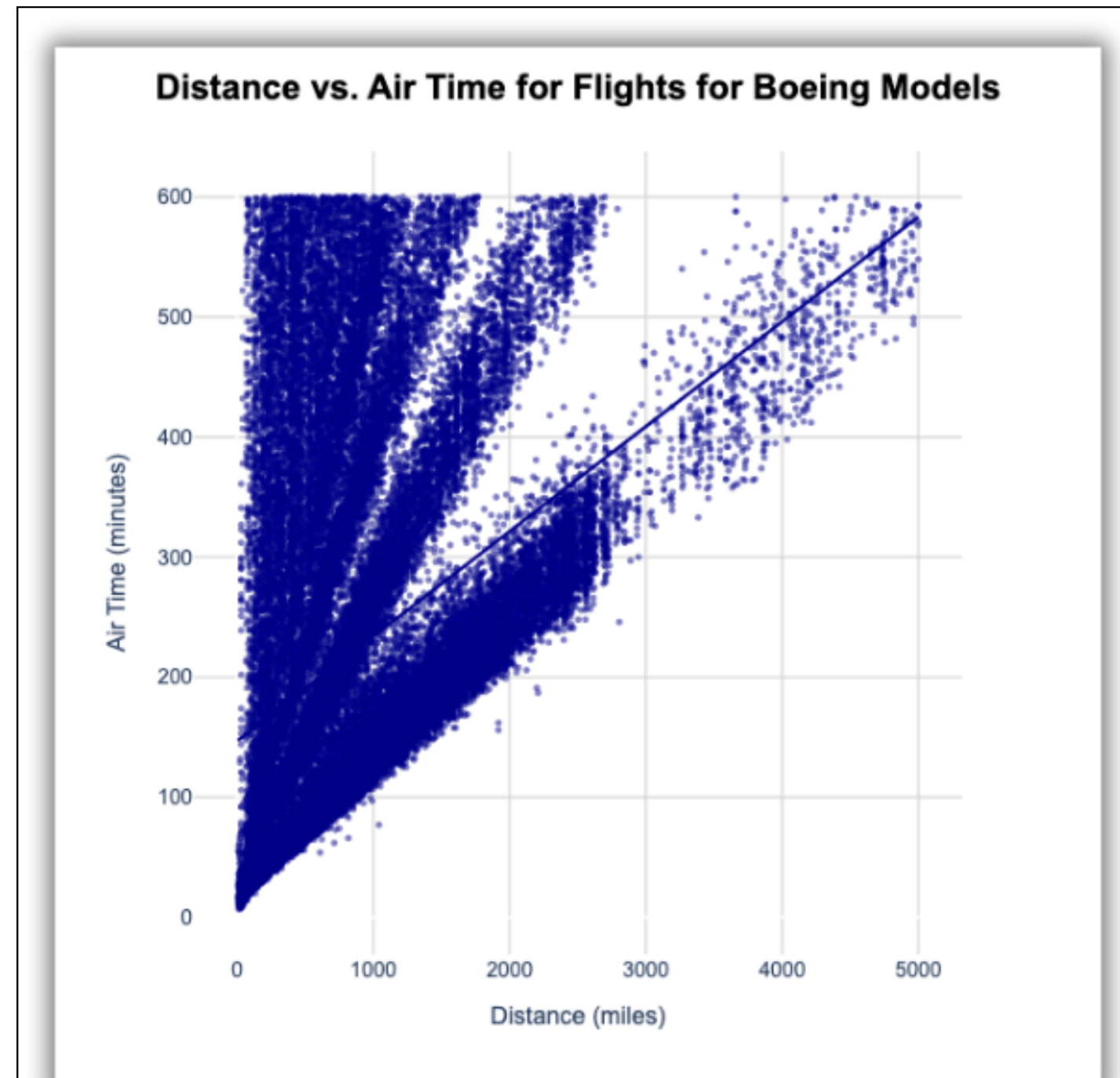
Boeing passenger volumes are highest in the larger states of California, Texas, Florida, Georgia, and Illinois



Boeing Passenger Volume by Major Hub State and Carrier



Southwest flies most in big states like California and Texas, while Delta, United, and American carry more passengers in key hubs like Georgia and Florida with fewer flights.



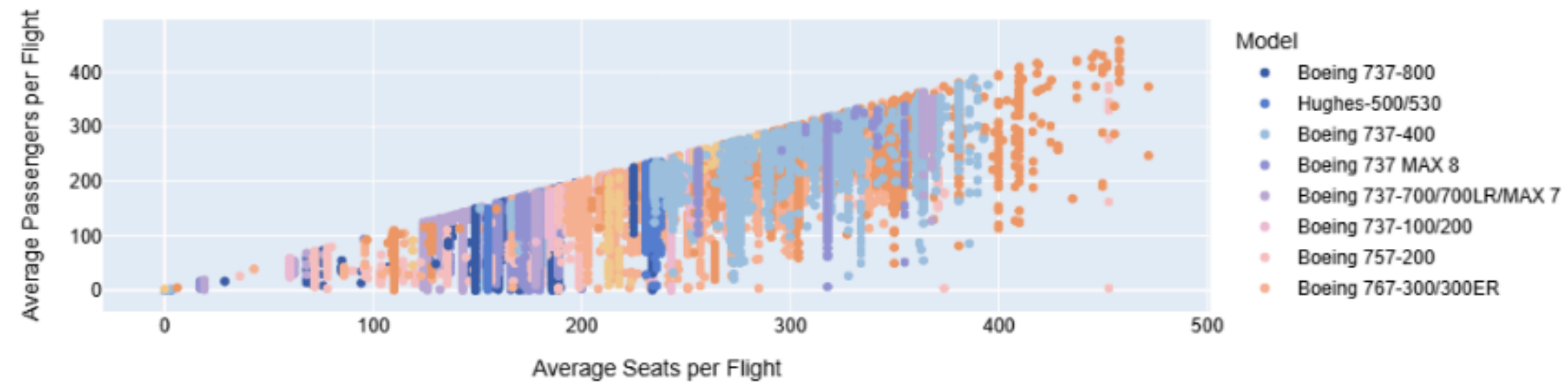
Air time rise with distance, with clusters showing flight types and outliers suggesting delays or inefficiencies. Some short flights take longer due to airport delays, weather, or indirect routes causing outliers in the plot.

### Passenger vs. Flight Volume (Boeing Models)

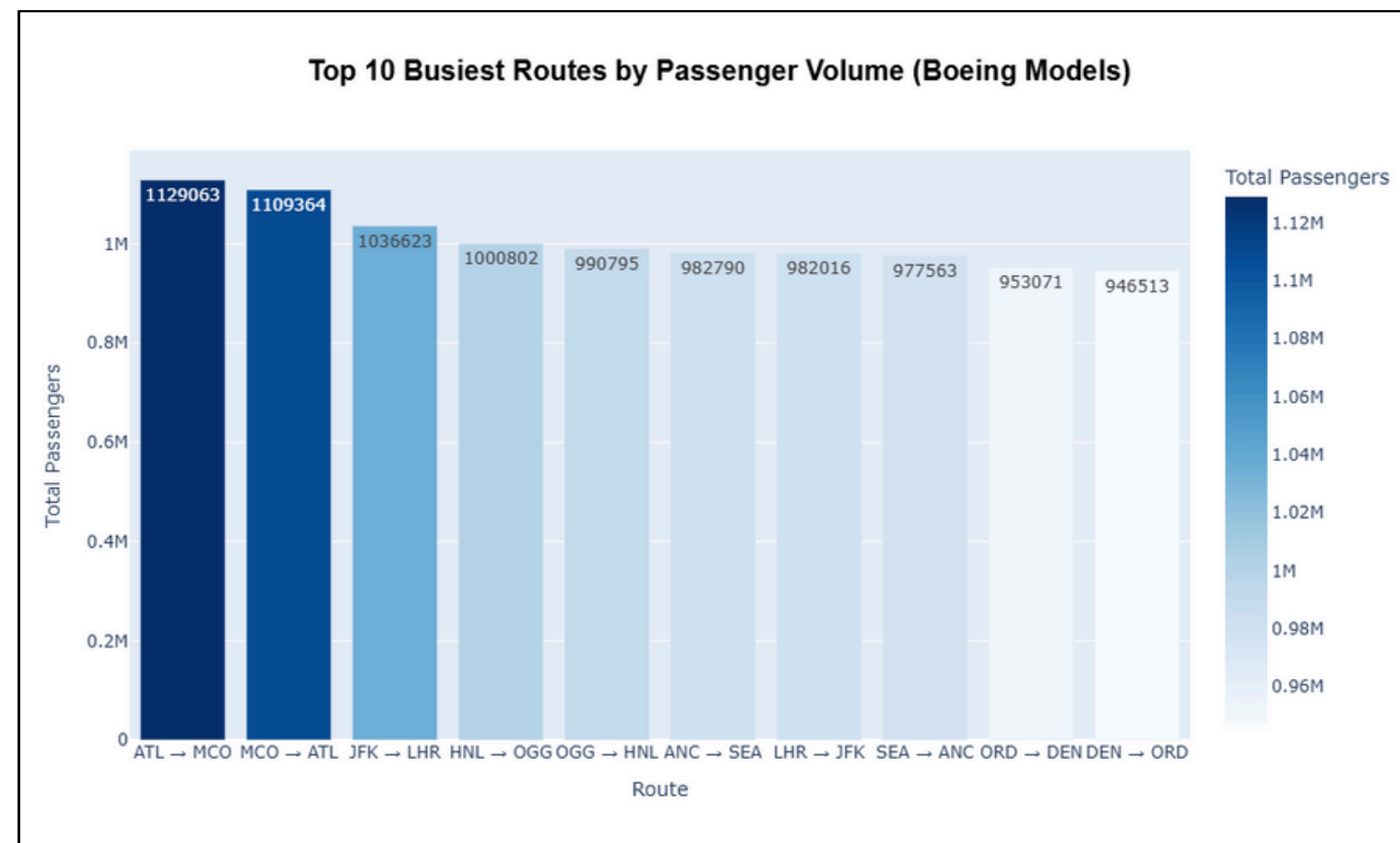


95% of routes with >400 flights and >40,000 passengers are Hawaiian inter-island (KOA, HNL, LIH, OGG); the rest show a balanced flight-passenger distribution.

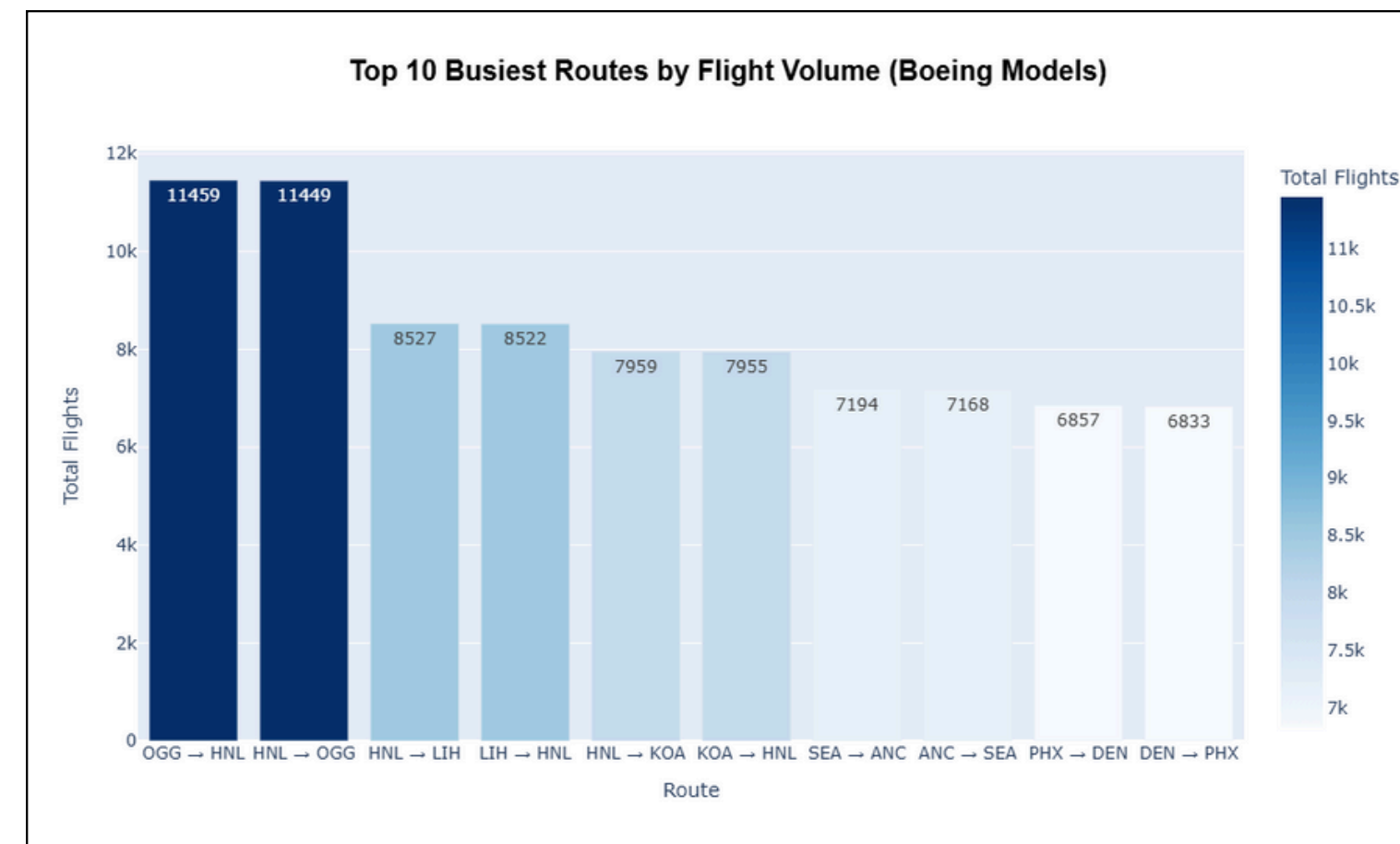
**Average Passengers vs. Seats per Flight by Aircraft Model**



Boeing models demonstrate a clear linear relationship, where an increase in seat capacity consistently leads to higher passenger counts



Boeing aircraft are busiest on high-traffic routes like ATL ↔ MCO and JFK ↔ LHR

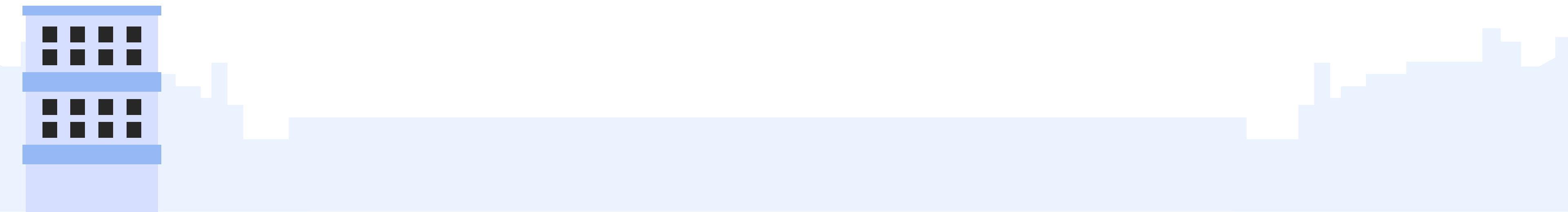


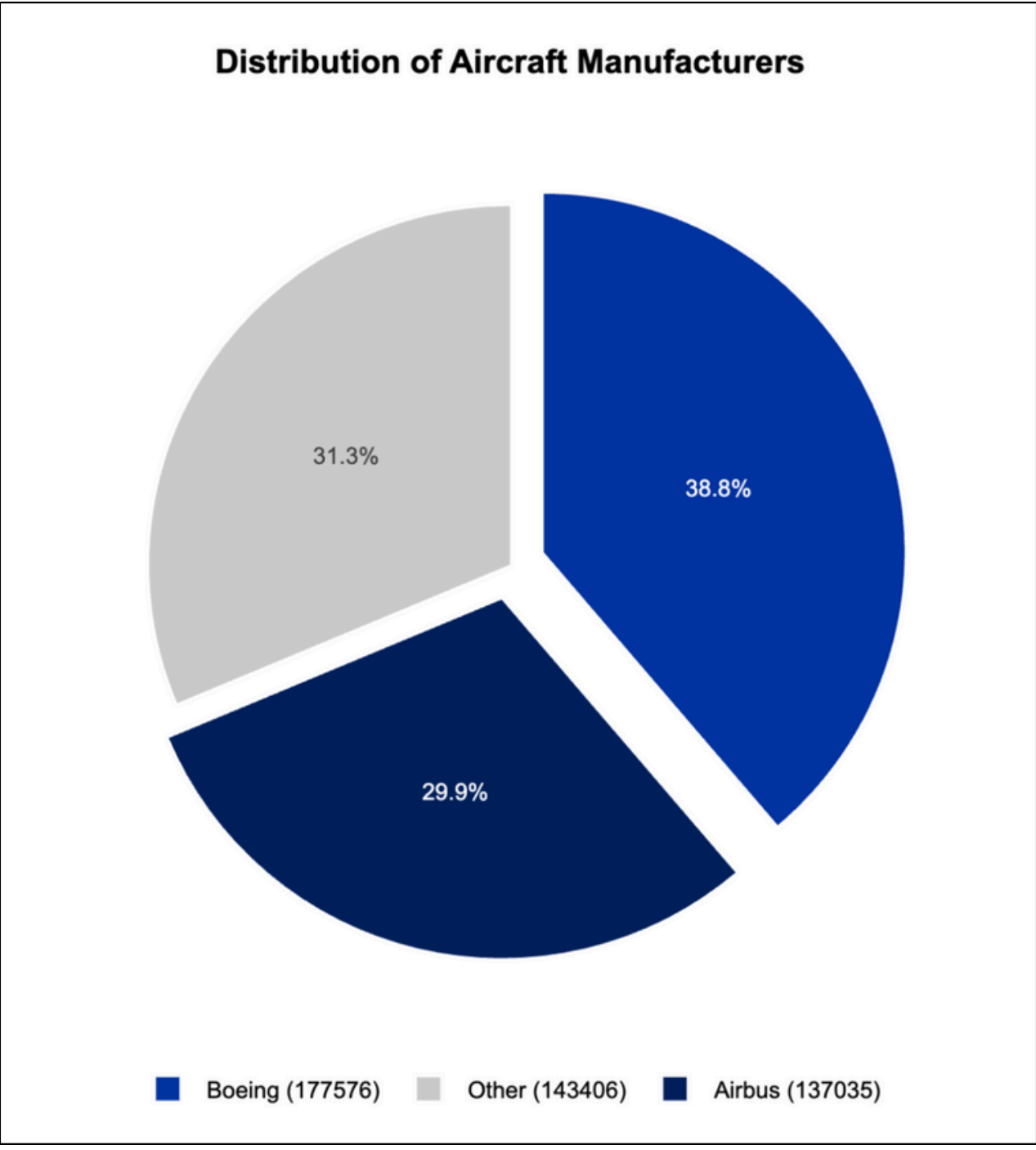
The OGG ↔ HNL route has the highest flight count, highlighting Boeing's dominance on Hawaii's inter-island routes



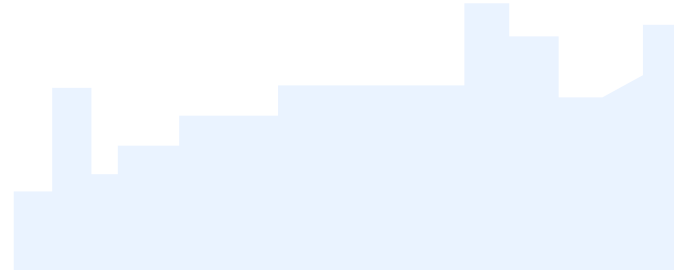
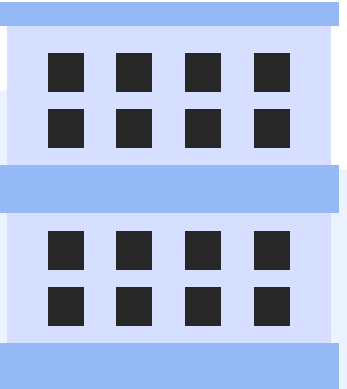


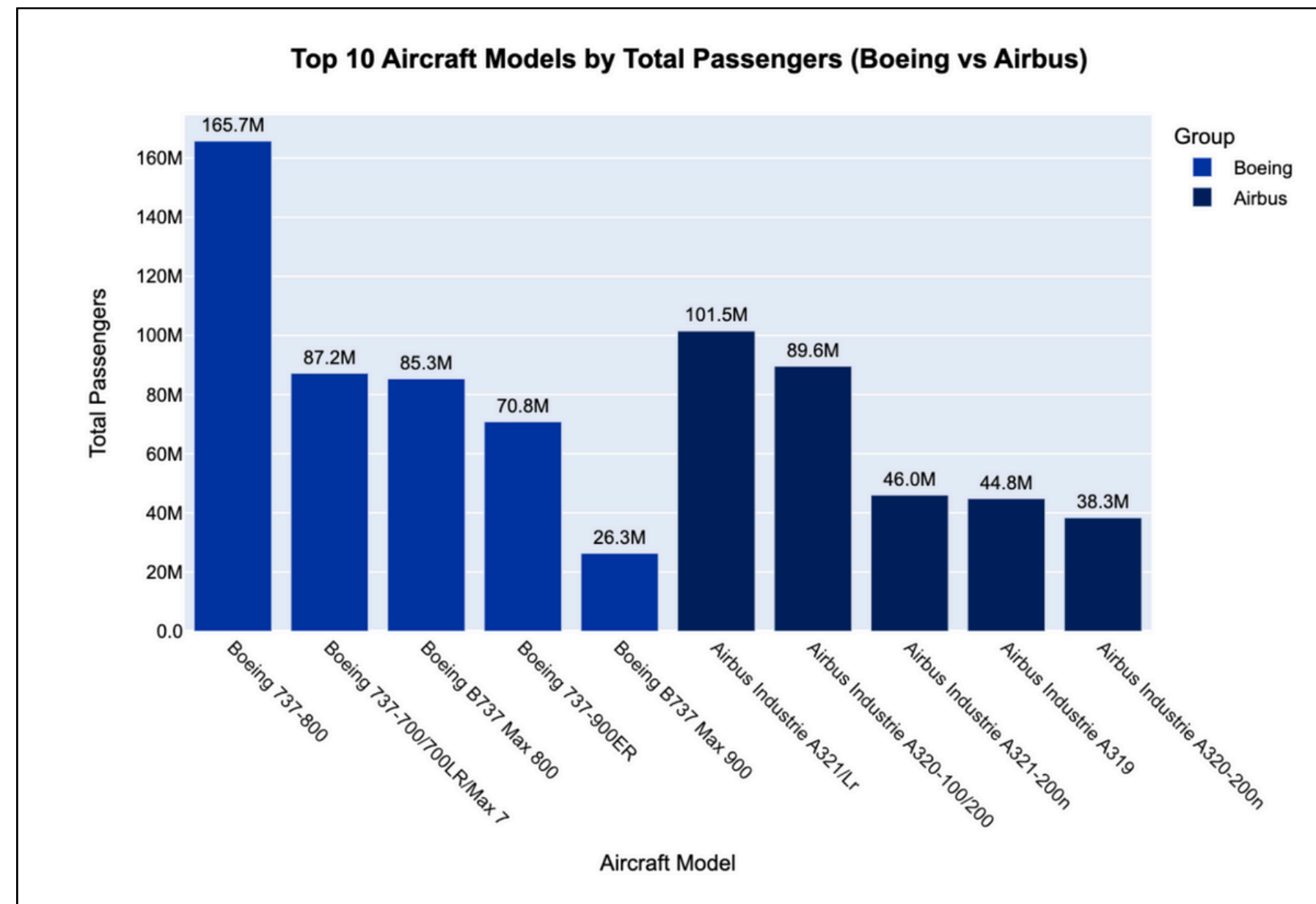
# ***Boeing vs Airbus***





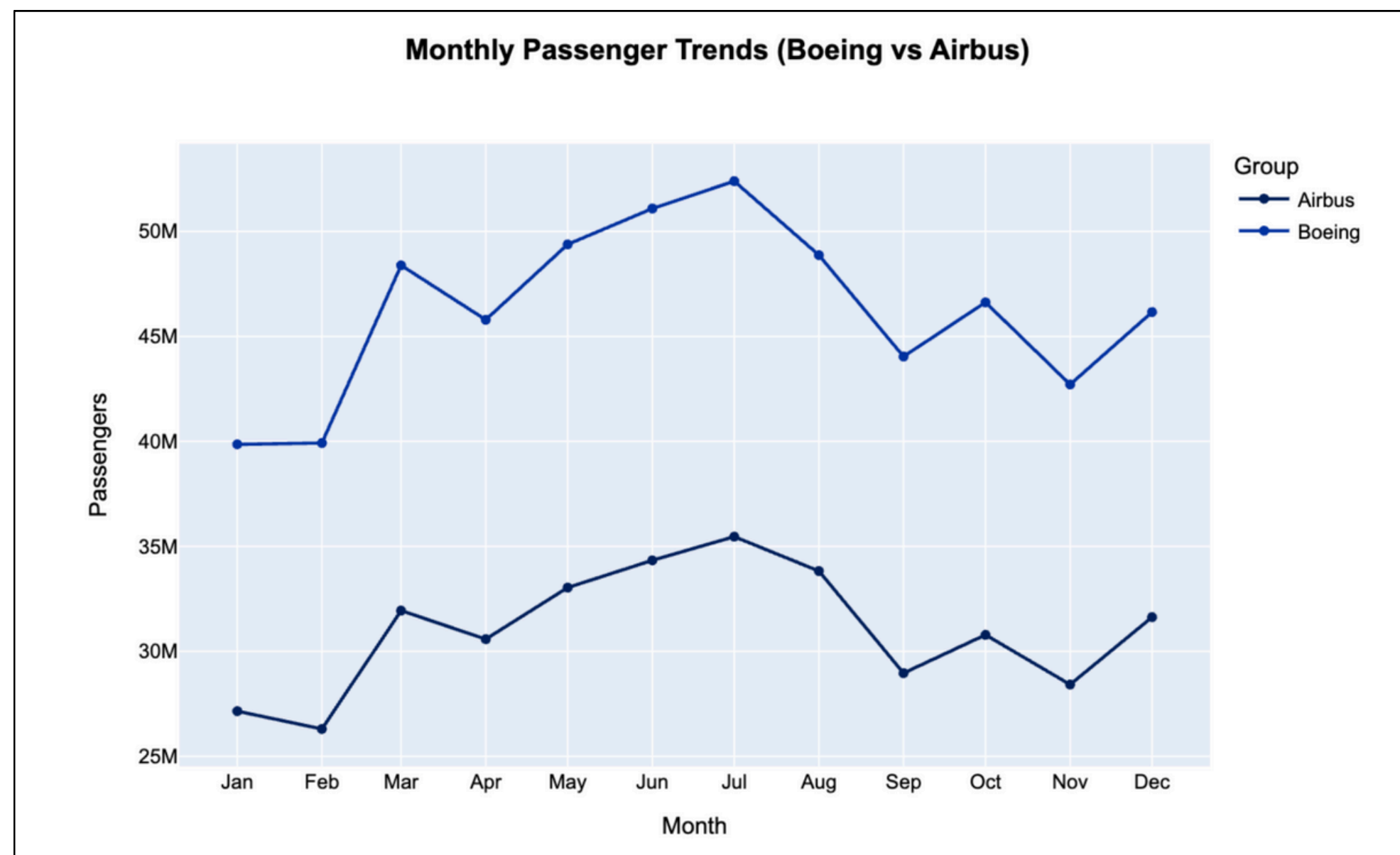
Out of 30 total manufacturers, Boeing/McDonnell-Douglas represents ~39% of the dataset and Airbus represents ~30%. The other 27 manufacturers make up the remaining 31%.



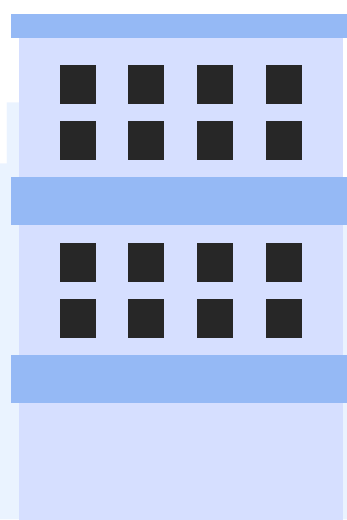


Boeing's top models are all from the 737 family, showing strong dominance, while Airbus's top five are all A320 variants, reflecting balanced popularity within the family.

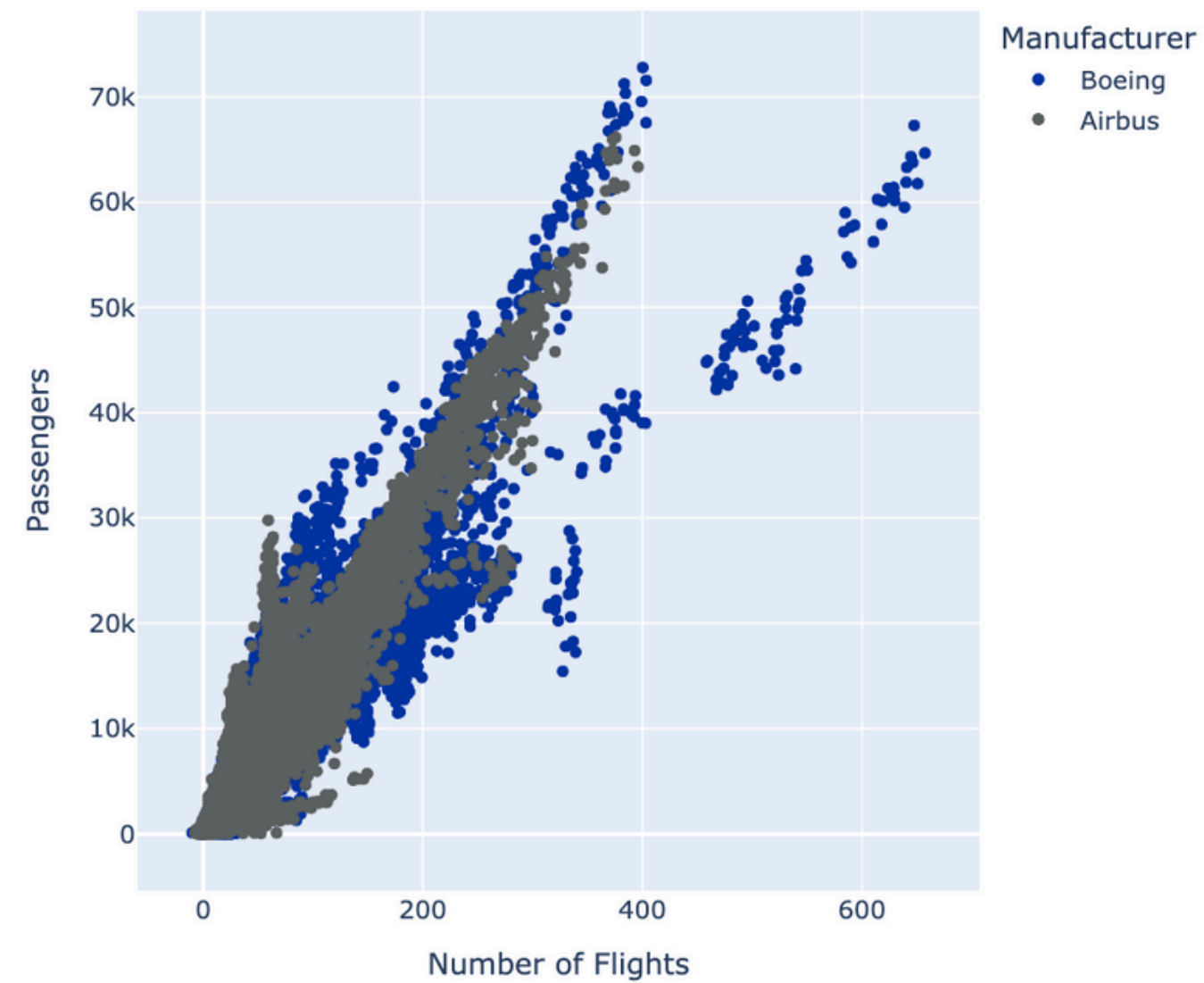




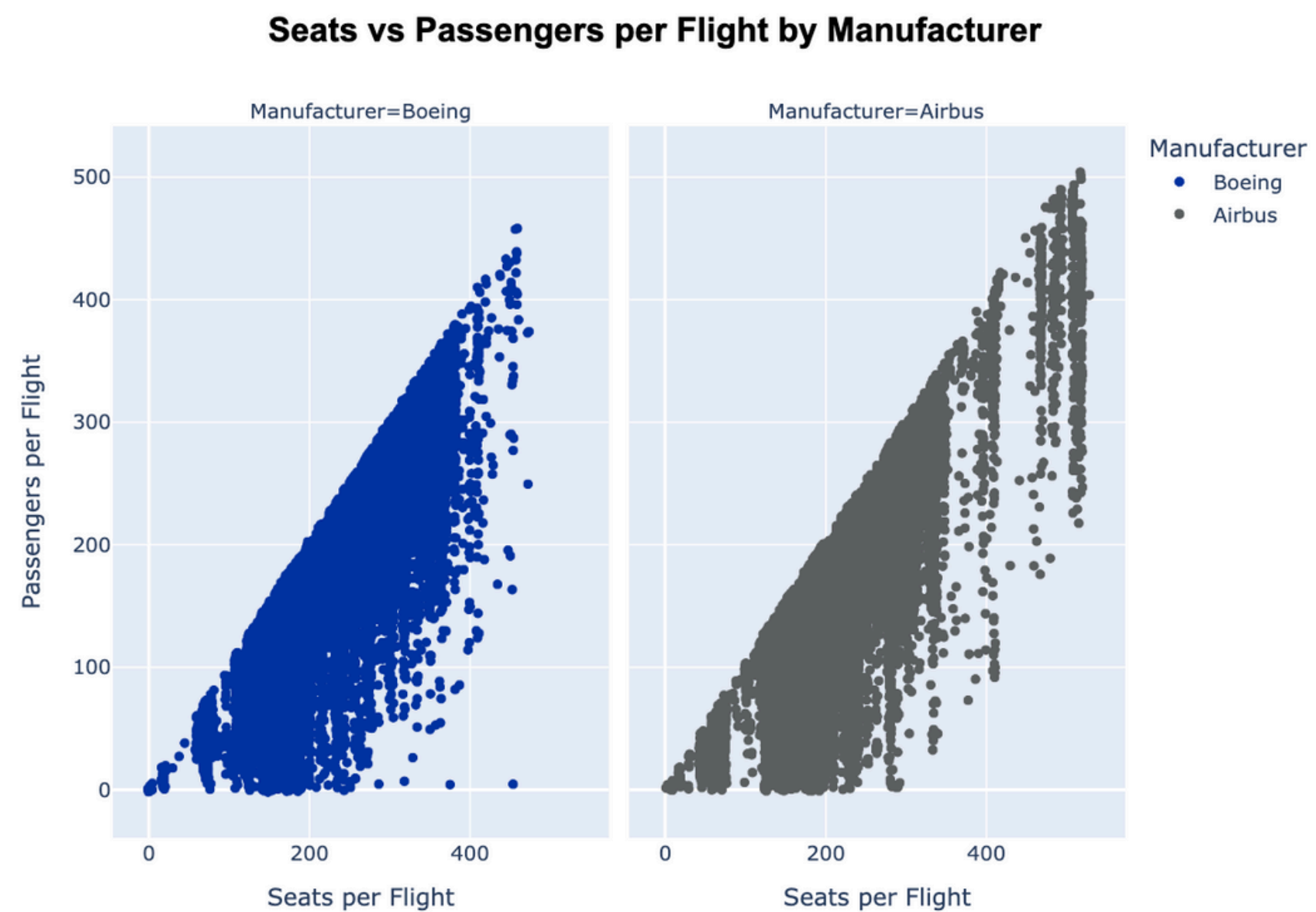
Both Boeing and Airbus follow the same monthly trend, with peaks in the summer and lows in February and November.



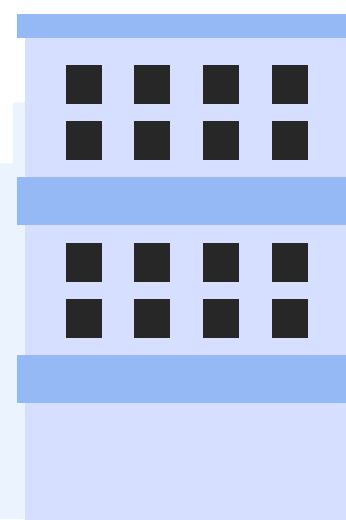
### Passengers vs Number of Flights by Manufacturer



Boeing carries more passengers than Airbus despite having fewer flights, indicating Boeing likely operates larger aircraft or more efficient routes.

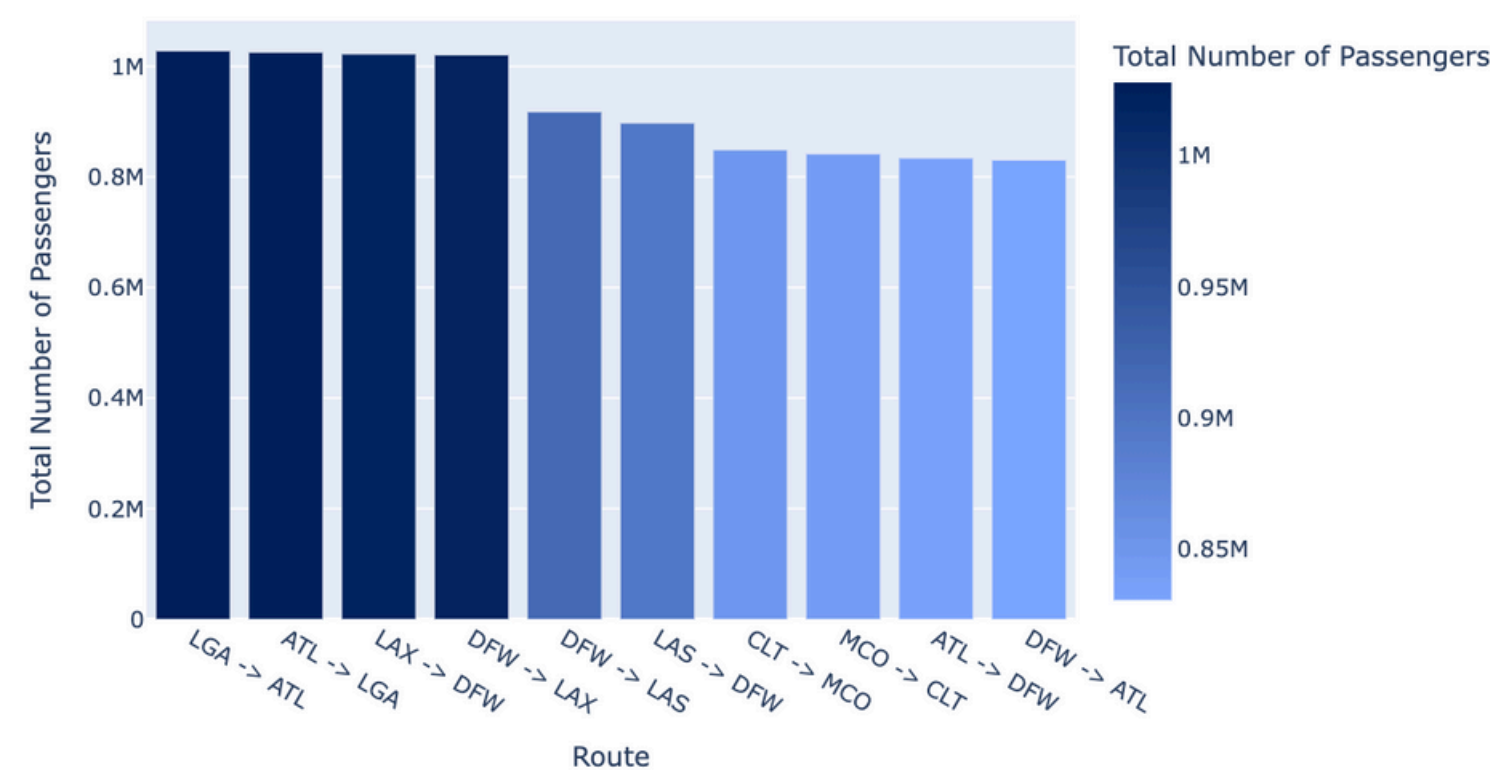


Boeing and Airbus share the same values, except Airbus has more for seat counts above 400 due to its larger planes.

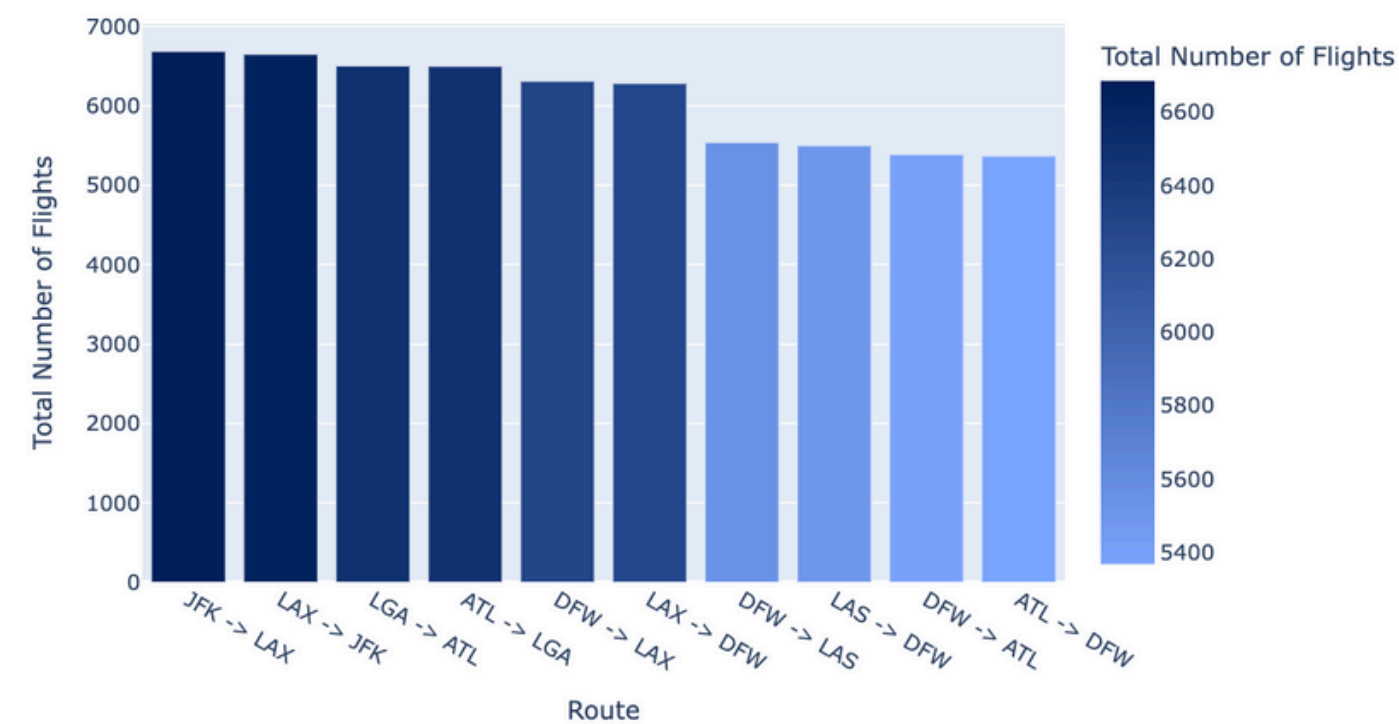




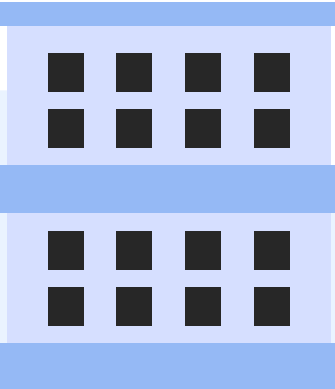
Top 10 Busiest Routes by Passenger Volume (Airbus Models)

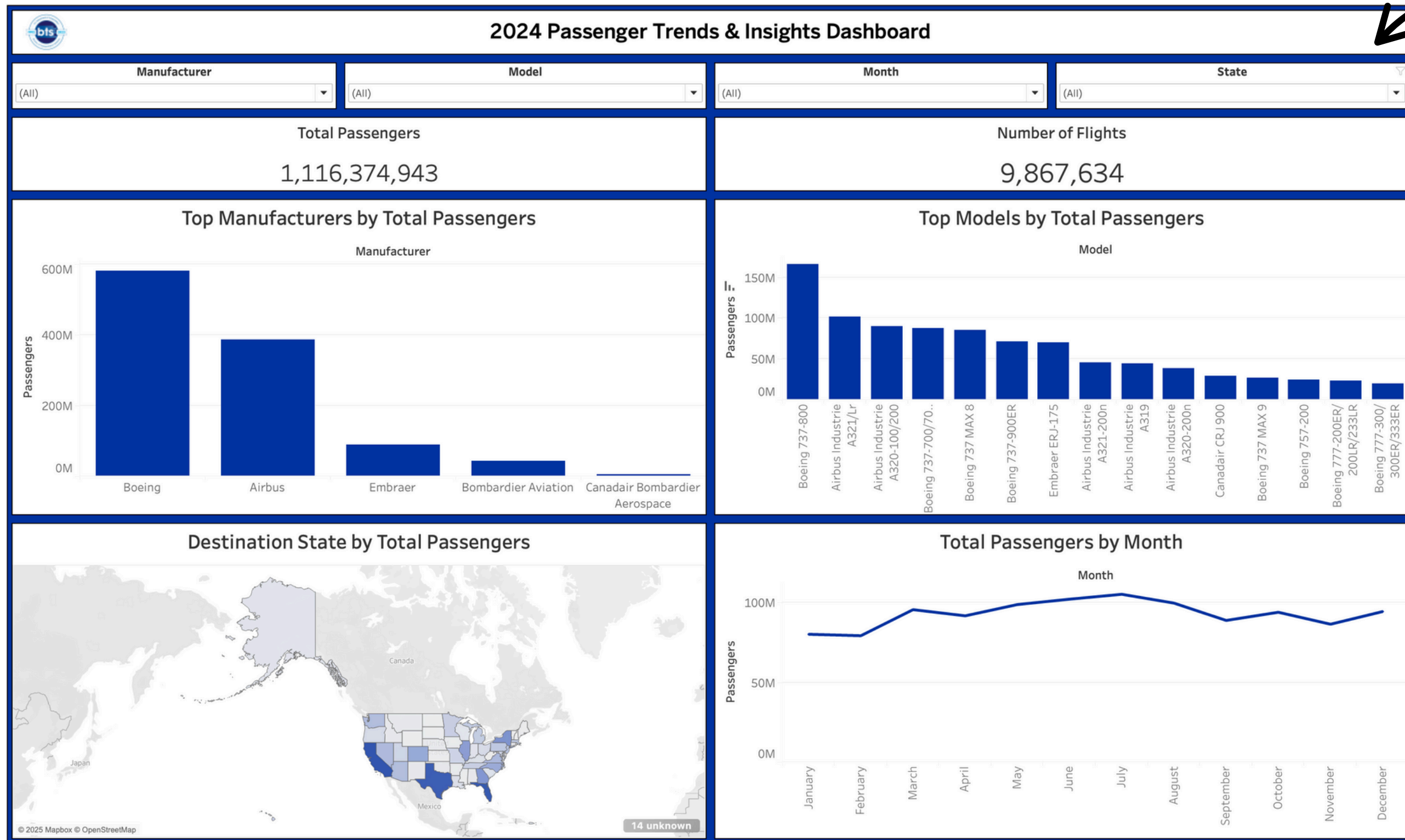


Top 10 Busiest Routes by Flight Volume (Airbus Models)



Routes like LGA → ATL and ATL → LGA carry the highest passenger volumes despite not having the most flights, indicating high aircraft occupancy or larger plane usage on these routes.





***Click for Dashboard***

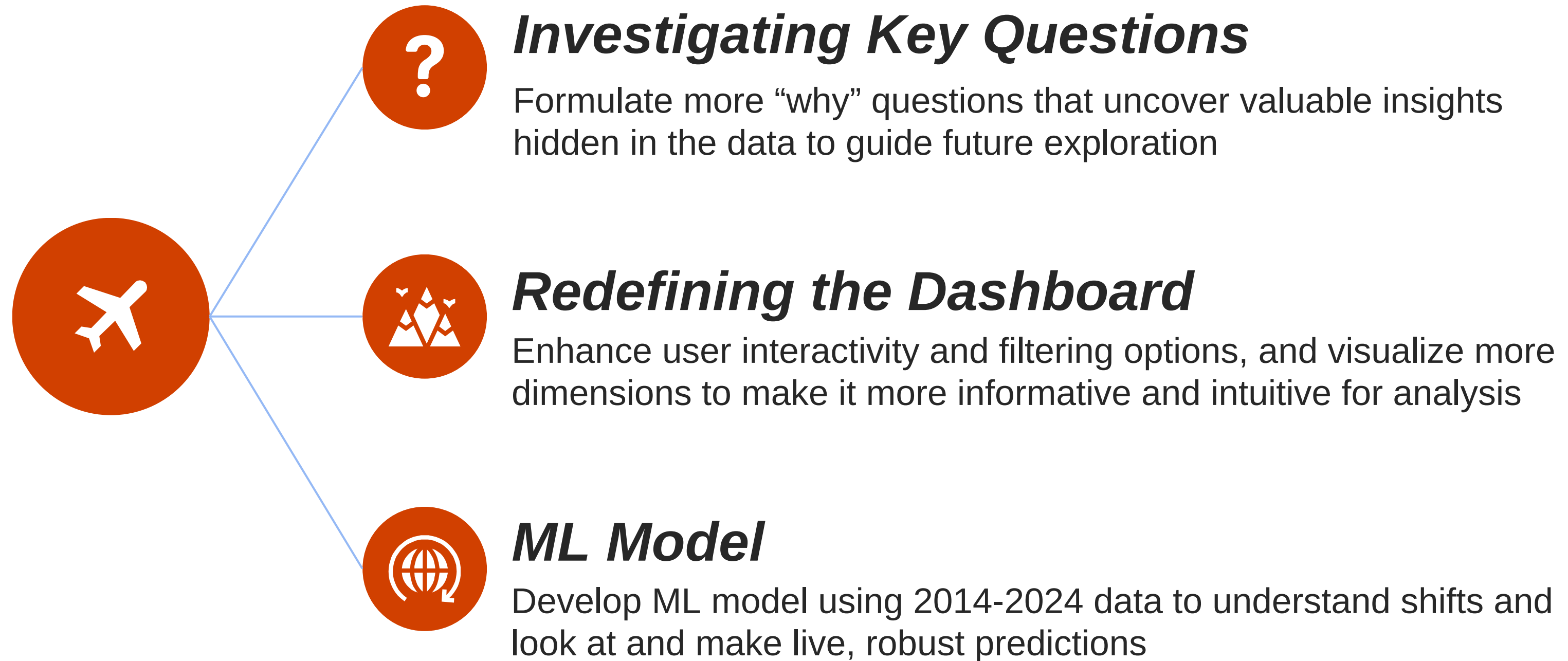


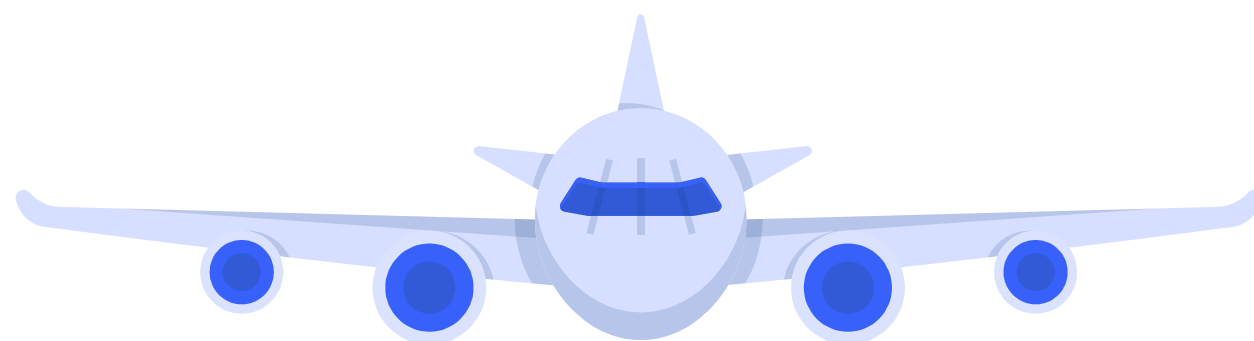


4

***Future Work***

# ***Future Work Directions***





*Thank You!*

*Any Questions?*

